ISEN - Junia

Module Numérique Responsable

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Audit Report: UGC Ticketing (Desktop)

TD: Eco-design of digital services

Group Members:
Amine EL BAKALI
Benjamin HUBERT
Cyprien KELMA
Nathan EUDELINE
Paul POUSSET
Marco FORDELONE

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1. 1. Framing the Audit (Scope)

1.1. 1.1. The Digital Service

Service Name: UGC Website (ugc.fr)
Platform: Desktop (PC)
Goal: Book a student ticket for a new movie at UGC Ciné Cité Lille.

1.2. 1.2. The Persona

Name: Student (-26 years old).
Constraint: Needs to book a specific session ("Film Nouveauté") for tonight. Has a password manager (good for auth) but is annoyed by complex navigation.

1.3. 1.3. The Functional Unit (FU)

> **Functional Unit:** [Booking 1 discounted ticket (-26) for a "New Release" movie at UGC Lille, including mandatory account login.]

1.4. 1.4. The Scenario (Path)

1. **Home:** Landing & Cookie Management.
2. **Movie Select:** Interaction with "Films Nouveautés" carousel.
3. **Cinema/Time:** Filter for UGC Lille, select evening session.
4. **Authentication: Mandatory** Login/Account Creation.
5. **Tariff:** Select "–26 ans" rate + Promo code check.
6. **Seat:** Select seat on HTML map.
7. **Upsell:** "Formules gourmandes" (Refusal).
8. **Recap:** Review & CGV validation.
9. **Payment:** Final checkout.

2. 2. Functional Analysis

2.1. 2.1. User Experience & Complexity

Step	Observation	Complexity / Pain Points
Home	Cookie Popup	Easy to reject cookies, but heavy page load.
Movie	Carousel	Carousel is **not scrollable**. Clicking poster opens preview; must click small text to select. Confusing UI.
Cinema	Filters	Localisation request is useless (doesn't auto-select). Cannot type city name (must pick from list).
Auth	Mandatory	**Blocker:** No Guest Checkout. Privacy policy is visible, but process is long.
Seat	HTML Map	Map is rendered fully in HTML (heavy DOM). Good for accessibility, bad for performance?
Upsell	Dedicated Page	Entire page load just to say "No" to food. Disruption of flow.

2.2. 2.2. C.O.E.U.R. Method Application

Analysis of features to optimize:

- **Eliminate (E):** The Upsell Page. Moving from Step 6 (Seat) to Step 8 (Recap) requires loading a heavy page (2.3 MB) for food. This should be a simple modal or integrated into the Tariff page.
- **Eliminate (E):** Mandatory Account. The biggest friction point.
- **Organize (O):** Cinema Selection. The city list is rigid. A simple search bar would be faster and reduce cognitive load.
- **Conceal (C):** Carousel Images. On the movie detail page, images for **other** movies are loaded but hidden in the non-scrollable carousel.

3. 3. Quantification of Impacts

3.1. 3.1. Environmental Performance Data

Page Name	EcoIndex	DOM	Reqs	Size (KB)
Home	F (23.1)	1819	84	3914
Movie	F (18.3)	2054	104	4266
Auth/Session	C (57.5)	279	79	3855
Tariff	C (53.6)	382	76	4099
Seat Sel.	C (64.1)	253	63	2513
Upsell	C (64.0)	268	63	2350
Recap	C (68.9)	178	57	2226
Payment (Est)	C (68.9)	178	57	2226
Total	**D (52.3)**	**5411**	**583**	**25449**

Table 1: GreenIT Analysis Results (Actual Data)

3.2. 3.2. Impact Equivalents

Total Journey Impact:

- **GHG Emissions:** 15.63 gCO2e
- **Water Consumption:** 23.44 cl
- **Data Transferred:** 24.85 MB

Real-world Equivalents:

- Travel: Equivalent to driving **71.7 meters** in a car.
- Water: Equivalent to **0.5 bottles** of water (50cl) per ticket bought.

3.3. 3.3. Annual Projection

Hypothesis: 15 Million visits/year.

- **Total CO2:** $15,000,000 \times 15.63\text{g} \approx 234.45$ tonnes CO2e
- **Total Water:** $15,000,000 \times 23.44$ cl $\approx 3,516\text{m}^3$ (approx 1.4 Olympic pools).

4. 4. Technical Analysis

4.1. 4.1. Comparison with Targets

DOM Elements:
Target: 600
Home/Movie: 2000 (Critical)

Page Weight:
Target: 1000 KB
Average: 3100 KB

Booking Funnel: 200 (Excellent)
Observation: Marketing pages are bloated; booking engine is visually simple but heavy in code.

Total: **24.85 MB**
Status: FAILED (Extremely heavy)

4.2. 4.2. Technical Diagnosis

Top Heaviest Resources:

- Images: Analysis shows images are not optimized (Home: 786KB potential gain, Movie: 958KB potential gain).
- Scripts: High number of inline stylesheets and scripts (120 inline blocks on Movie page).
- Legacy Stack: The `.action` URLs in the booking funnel (Steps 4-9) suggest a legacy Java/Struts framework. While the DOM is light, the server response includes heavy headers and uncompressed assets (4MB per page for simple forms).

Specific Issue - Upsell Page: The "Upsell" page (Step 7) downloads 2.35 MB of data. If the user refuses food, this is 100% digital waste.

5. 5. Recommendations

5.1. 5.1. Level 1: Technical Optimizations

- **Action:** Image Optimization. The audit tool identified 1.7 MB of immediate savings just by compressing images on Home and Movie pages.
- **Action:** Cache Control. The logs show "Add expires headers" is missing for dynamic resources.
- **Action:** Minification. The booking funnel has unminified CSS/JS.

5.2. 5.2. Level 2: Architecture

- **Action:** Lazy Loading. The Movie page loads 4.2 MB. Much of this is the "Films Nouveautés" carousel. Images should only load when the user scrolls/clicks.
- **Action:** Merge Steps. Combine "Tariff" and "Upsell". Show food options as small add-ons on the Tariff page instead of a dedicated full-page load.

5.3. 5.3. Level 3: Functional Innovation

- **Action:** Guest Checkout. Remove mandatory account creation.
- **Action:** Smart Location. If the user allows geolocation (as requested in Step 3), automatically sort the cinema list instead of making them search manually.

5.4. 5.4. Level 4: Radical Innovation

- **Concept:** "UGC Lite" Text Mode.
- A simple interface for repeat users: "Lille > Tonight > Gladiator II > Pay".
- Bypasses the Home and Movie pages entirely (saving 8MB per transaction).

6. Conclusion

The UGC audit reveals a "Tale of Two Sites". The Marketing pages (Home, Movie) are extremely heavy (Grade F), bloated with unoptimized images and DOM. The Booking Engine (Auth to Payment) is visually simple but technically heavy (Grade C), with pages averaging 3MB despite low visual complexity. The total footprint of **24.85 MB** for a single ticket is excessive. By removing the mandatory account and the dedicated Upsell page, we could reduce the journey's impact by nearly 20%.